REFERENCE AGREEMENT IS NOT ACOUSTIC CORRECTNESS: EVALUATING LARGE AUDIO LANGUAGE MODEL CAPTIONS AGAINST HUMAN REFERENCES ON CLOTHO

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ABSTRACT

Large audio language models (LALMs) describe sound recordings fluently, but their captions are usually judged by how closely they agree with human references. We evaluate captions from four LALMs, GPT-Audio-1.5, Gemini 3.8 Flash, Qwen2.5-Omni-7B and Audio Flamingo 3, on the complete Clotho v2.1 evaluation split (1,045 recordings, 5,225 human captions) with twelve reference-based and two audio-aware metrics. A matched leave-one-out procedure scores held-out human captions against the same four references as the models. The two open-weight models score highest on every reference-based metric, yet all four models remain below the human baseline on each of them; only Audio Flamingo 3 on CLAP audio–text similarity is not significantly different from it. Model captions are 1.4–1.9 times longer than human captions, and longer captions score lower on overlap metrics. A single-reviewer audit of 200 sampled captions against the audio reverses the model order: GPT-Audio-1.5, ranked third or fourth by the text metrics, was rated accurate for 47 of 50 recordings, and 56 of the 64 captions that disagree most with the references had no major error. No evaluated metric showed significantly above-chance within-recording concordance with the audio ratings in this sample. Reference agreement alone is therefore insufficient evidence of acoustic correctness.

***Index Terms*—** Audio captioning, large audio language models, evaluation metrics, Clotho, human references

1. INTRODUCTION

Automated audio captioning (AAC) describes the sound sources, events and acoustic characteristics of a recording in natural language [1]. Applications include accessibility, multimedia retrieval and soundscape analysis. Large audio language models (LALMs) now perform this task without task-specific training, and produce longer and more detailed descriptions than captioning systems trained on AAC datasets [2, 3, 4, 5].

AAC systems are evaluated by comparing their captions with human references. Metrics such as BLEU [6], METEOR [7], CIDEr [8] and SPIDEr [9] measure the lexical or semantic overlap between a caption and the references. A recording, however, often contains several sounds, and different listeners describe different aspects of it. A caption may therefore describe an audible event that no reference mentions and receive a low score, or overlap well with the references while misrepresenting the audio. The difference matters more for LALMs, whose captions differ from the references in length and style.

In this paper we compare captions from four LALMs with the human references of Clotho v2.1 [10]. We address three research questions:

1. **RQ1: How closely do model captions agree with the five human reference captions?** We score each model's caption against all five references of its recording with twelve reference-based metrics and two audio-aware metrics, and compare the models.
2. **RQ2: How do model captions compare with held-out human captions in metric scores and caption length?** We score each human caption against the other four references and each model caption against the same four, so that model and human scores are directly comparable. We also compare caption lengths.
3. **RQ3: What patterns of agreement, additional detail, omission and mismatch appear in model captions?** We review a random sample of 200 model captions first against the references and then against the audio, and test whether the metrics follow either review.

We find that the reference-based metrics agree with each other and with the reference-only review, but we find no evidence that they follow correctness against the audio. The model with the most accurate captions in the audio review ranks third or fourth on every text metric. Section 2 reviews related work, Section 3 describes the methodology, Section 4 presents the results for RQ1–RQ3, Section 5 discusses their implications, and Section 6 states the limitations.

2. RELATED WORK

2.1. Automated audio captioning

AAC research relies on two crowdsourced datasets, AudioCaps [11] and Clotho [10], and has been driven by the DCASE challenge since 2020 [1]. AudioCaps provides one caption per training recording and five per test recording, drawn from AudioSet. Clotho provides five captions for each of its 15–30 s recordings, collected with a three-step protocol in which annotators write captions, other annotators correct or rephrase them, and the five captions rated highest for accuracy and fluency are kept [12]. Because both datasets are small, recent systems are pre-trained on larger, weakly labelled caption collections such as WavCaps [13]. Human references are therefore short, single sentences that describe the most salient sounds, and the style of these references shapes both training and evaluation.

2.2. Large audio language models

LALMs couple a pre-trained audio encoder with a large language model and are trained to follow free-form instructions about audio. Pengi [14] framed audio tasks as text generation, LTU [2] and SALMONN [3] added open-ended question answering and reasoning, and Qwen-Audio [15] scaled multi-task pre-training across speech, sound and music. Recent models handle audio within general multimodal assistants, such as Qwen2.5-Omni [4], or are trained specifically for audio understanding and reasoning, such as Audio Flamingo 3 [5]. These models are usually evaluated on captioning benchmarks with the same reference-based metrics as task-specific systems, although their captions are longer and more detailed than

the references. LALMs can also describe sounds that are not present in the recording [16], which reference-based scores alone cannot reveal.

2.3. Audio caption evaluation metrics

Most AAC metrics were borrowed from image captioning and machine translation: BLEU [6], METEOR [7] and CIDEr [8] measure n-gram overlap, SPICE [17] compares scene graphs parsed from the captions, and SPIDEr [9] averages CIDEr and SPICE. Zhou et al. [18] showed that these metrics agree poorly with human judgements of audio captions and proposed FENSE, which combines Sentence-BERT similarity with a penalty for fluency errors. Other work scores several candidate captions jointly [19], measures semantic agreement of sound descriptors [20], compares contextual token embeddings with BERTScore [21], or asks a large language model to judge captions [22]. All of these metrics compare a caption with human references. Audio-aware metrics use the recording instead: CLAP [23, 24] embeds audio and text in a joint space, so a caption can be scored against the audio, as CLIPScore [25] does for images, and MACE [26] combines CLAP audio–text and text–text similarity with a fluency penalty.

2.4. Reference agreement and acoustic grounding

Scores against human references are hard to interpret without knowing how well humans agree with each other. In image captioning, human captions scored against the remaining references give such a reference point [27], and object hallucination is measured by checking caption content against the image rather than against the references [28]. Studies of AAC metrics have compared metric scores with human preferences between caption pairs, judged on description accuracy and fluency [18, 26]; they do not rate each caption against a correctness rubric or check the details that no reference supports. This work combines both ideas for LALM audio captions: we calibrate model scores with a matched human leave-one-out baseline, and we verify each sampled caption, and each detail that no reference supports, against the audio.

3. METHODOLOGY

3.1. Data and models

We use the complete evaluation split of Clotho v2.1 [10]: 1,045 recordings of 15–30 s (6.5 h in total), each with five human captions of 8–20 words (5,225 captions). We evaluate two API models, GPT-Audio-1.5 (`gpt-audio-1.5`) and Gemini 3.8 Flash (`gemini-3.8-flash`), and two open-weight models, Qwen2.5-Omni-7B [4] and Audio Flamingo 3 [5], at pinned checkpoint revisions. Every model received a single user turn with the full recording followed by the same instruction, shown in the prompt template below. No references were shown, no length was imposed, and no model-specific prompt tuning was done; only the provider-specific encoding of the audio differed.

Prompt template (one user turn, identical for all models)

`[audio]` *full recording, 15–30 s, no cropping*
`[text]` Describe the sounds heard in this audio recording in one clear sentence that mentions the main sound sources, audible events, and distinguishing acoustic characteristics. Do not speculate about sources or events that are not supported by the audio. Respond with the sentence only, as plain text, without JSON, lists, labels or other formatting.

Decoding used temperature 0, top-p 1 and a seed of 42 where supported; Gemini 3.8 Flash ran at its lowest thinking level (`low`), as thinking cannot be disabled. This gave one caption per model and recording (4,180 in total). GPT-Audio-1.5 sometimes answered in JSON despite the instruction: 85 captions were plain sentences wrapped in braces and were unwrapped, and 21 malformed captions were scored as returned. Excluding the 21 affected recordings changes no model order and no significance decision. The API models are not deterministic at temperature 0, so all results are conditional on the released captions. Checkpoint hashes, API run dates and library versions are listed in the repository.

3.2. Metrics and human leave-one-out calibration

The reference-based metrics are BLEU-1 to BLEU-4, METEOR, CIDEr-D, SPICE [17], SPIDEr, SPIDEr-FL, FENSE, and the raw and baseline-rescaled F1 of BERTScore, computed on PTB-tokenized text with the aac-metrics package [29]. The audio-aware metrics are CLAP similarity, the cosine between the audio and caption embeddings of MS-CLAP 2023 [24], and MACE [26]. CLAP embeds each clip whole, as the duration-weighted mean of 7 s chunks. BERTScore uses the last layer of RoBERTa-large without IDF weighting. We compute it with our own batching code, because torchmetrics 1.9.0, which aac-metrics calls, restores length-sorted embeddings in the wrong order in batched calls and thus scores captions against other items' references; the repository includes the corrected code and a test.

For RQ2, let $H = \{h_1, \ldots, h_5\}$ be the human captions of a recording and m_k the caption of model k. For each position j, the held-out caption h_j and m_k are both scored against $R_{-j} = H \setminus \{h_j\}$. The five scores are averaged per recording; corpus-level metrics are computed once per position and averaged. Because CIDEr-D, and hence SPIDEr and SPIDEr-FL, depend on the reference set, five-reference (RQ1) and four-reference (RQ2) scores are never compared with each other. Clotho's protocol keeps original and rephrased captions without deduplication, which is consistent with the near-paraphrase pairs we observe: 33% of held-out captions share a word-set Jaccard overlap of at least 0.6 with one of their references. We therefore repeat the comparison without these captions.

3.3. Caption length and qualitative review

Caption length is the number of whitespace-separated words. We relate it to each score with Spearman correlations over captions. For RQ3, we drew 50 recordings at random (seed 42), independently of the scores, giving 200 model captions. A single reviewer labelled all captions and listened to all recordings by hand; no AI system assigned or suggested labels. Captions were shown blind to model identity, under per-recording random labels. The reviewer first compared each caption with the five references only and assigned one of six agreement categories: (1) strong agreement, (2) mostly aligned, (3) mostly aligned with additional details, (4) partially aligned with missing details, (5) partially aligned with additional details, and (6) weak agreement or source/event mismatch. Phrases supported by no reference were recorded. After these labels were final, the reviewer listened to each recording and rated each caption as (1) accurate, where omitting a secondary sound alone does not lower the rating, (2) minor errors: principal source and event correct, but a wrong attribute or order or an unsupported secondary detail, or (3) major error: principal source or event wrong or invented, or the dominant

sound omitted. The reviewer also judged each recorded phrase as supported, unsupported or unverifiable by the audio.

3.4. Statistical analysis

The recording is the unit of analysis, and every comparison is paired. We test mean differences with two-sided sign-flip permutation tests (10,000 permutations) and report 95% percentile bootstrap intervals over recordings (10,000 resamples). Holm correction is applied within predefined families: 84 model-pair tests for RQ1 and 56 model-versus-human tests for RQ2. We report the paired effect size d_z, because with 1,045 recordings very small differences are significant. Ranking stability is the share of bootstrap resamples that reproduce the observed model order. To relate the metrics to the audio review, we count, over pairs of captions of the same recording with different audio ratings, how often a metric scores the better-rated caption higher (concordance; ties count one half; 0.5 is chance). Its bootstrap intervals and sign-flip tests resample or flip whole recordings, and Holm correction runs over the ten metrics. Review proportions carry Wilson 95% intervals.

4. RESULTS

4.1. RQ1: Agreement with five references

Table 1 (left) shows that Audio Flamingo 3 scores highest and Qwen2.5-Omni-7B second on every reference-based metric. The two API models share the third and fourth places: for example, CIDEr-D is 0.191 and 0.164 for the open-weight models and 0.064 and 0.048 for Gemini 3.8 Flash and GPT-Audio-1.5. The order is stable: at least 99% of bootstrap resamples reproduce it for BLEU-1, BLEU-2, BLEU-4, METEOR, CIDEr-D, SPICE, FENSE and both BERTScores. The exceptions are BLEU-3 (87%) and SPIDEr (55%), where GPT-Audio-1.5 and Gemini 3.8 Flash are tied. After Holm correction, 76 of the 84 model-pair differences are significant. Absolute agreement is low: CIDEr-D ranges from 0.05 to 0.19.

The audio-aware metrics order the models differently. CLAP similarity and MACE score GPT-Audio-1.5 close to Qwen2.5-Omni-7B (0.466 and 0.462; 0.573 and 0.575), and Gemini 3.8 Flash is last on both. Per caption, the text metrics correlate with each other (Spearman ρ = 0.41–0.88), whereas CLAP and MACE correlate 0.89 with each other but only 0.28–0.67 with the text metrics.

4.2. RQ2: Comparison with held-out human captions

Under the same four references, every model is below the human baseline on every reference-based metric, with large effects (d_z from -0.48 to -2.55; Table 1, right). 55 of the 56 comparisons are significant after Holm correction. The exception is CLAP similarity, where Audio Flamingo 3 is not significantly different from the human captions ($+0.006$, $d_z = 0.06$); when held-out captions with a near-paraphrase reference are removed, Audio Flamingo 3 is significantly above the human baseline on CLAP similarity (uncorrected). The near-paraphrase pairs inflate the human baseline: without them, human CIDEr-D falls from 0.91 to 0.46 and rescaled BERTScore from 0.60 to 0.47. Even so, every model remains significantly below human on all reference-based metrics.

Model captions are 1.45 (Gemini 3.8 Flash) to 1.85 (GPT-Audio-1.5) times as long as human captions, and 92% of GPT-Audio-1.5's captions are longer than every reference of their recording. For the models, length correlates negatively with the overlap metrics: from -0.09 (Gemini 3.8 Flash) to -0.63 (Audio Flamingo 3) for CIDEr-D, and from -0.19 to -0.52 for rescaled BERTScore. CLAP similarity and MACE are almost independent of length, except for Audio Flamingo 3 (-0.11 and -0.17), and human captions show near-zero correlations throughout.

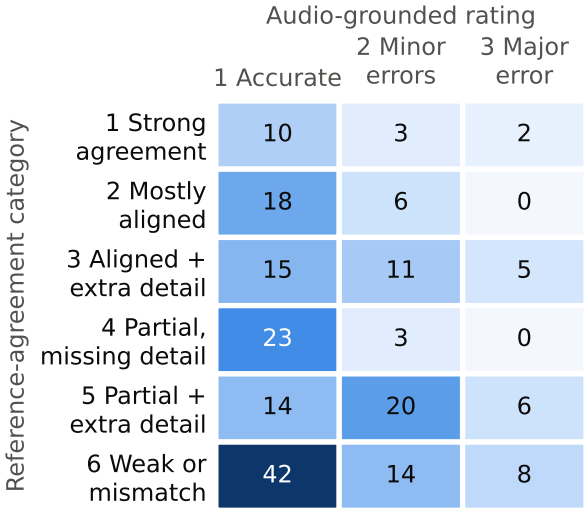


Fig. 1. Reference-agreement category (rows, judged against the references only) by audio-grounded rating (columns, judged against the recording) for the 200 sampled captions of all four models.

4.3. RQ3: Agreement patterns and acoustic correctness

The sample was similar to the full set on the measured summary statistics (standardized differences of at most 0.37). Against the references, Audio Flamingo 3 and Qwen2.5-Omni-7B agree most often with the main content and Gemini 3.8 Flash least often (Table 2). GPT-Audio-1.5's most frequent deviation is omission: it has the most captions in category 4.

Against the audio, the order changes. GPT-Audio-1.5 was rated accurate for 47 of 50 recordings (94%, 95% CI 84–98%) with no major error, while the other three models were rated accurate for 23 to 27 (46–54%, intervals within 33–67%) and had 5 to 9 major errors. The most direct evidence of missing reference content is phrase-level: of the 262 phrases that no reference supports, 64% [58, 70] are audible in the recording, 28% are not and 8% cannot be verified. Unsupported phrases occur in 26 of Gemini 3.8 Flash's captions, 19 of Qwen2.5-Omni-7B's and 15 of Audio Flamingo 3's, but in only one of GPT-Audio-1.5's. At the caption level (Fig. 1), only 8 of the 64 captions in category 6 have a major error, and 23 of the 26 category-4 captions are rated accurate. Disagreement with the references is therefore mostly not an acoustic error in this review; incomplete references are the likeliest explanation, although reference errors, ambiguous sounds and reviewer judgement may contribute.

The metrics follow the reference review, not the audio review. Kendall's τ between each metric's model order and the reference-review order is 0.67–1.00, but with the audio-review order it is 0–0.33 for every metric except CLAP similarity (0.67). Mean scores fall from category 1 to category 6 on every metric (e.g. FENSE 0.68 to 0.28), but are flat or reversed across audio ratings (rescaled BERTScore 0.29, 0.31 and 0.34 for ratings 1, 2 and 3). Within a recording, across 156 caption pairs from 42 recordings with different

Table 1. Mean per-recording scores. Left: model captions against all five references (RQ1); bold marks the highest model. Right: held-out human captions against four references, and the paired effect size d_z of each model minus the human baseline under the same four references (RQ2). Every difference is Holm-significant except the one marked †. The last row gives mean caption length and d_z of model length minus the mean reference length.

	Five references				Human	Model − human, d_z (four references)			
Metric	GPT	Gemini	Qwen	AF3	(four ref.)	GPT	Gemini	Qwen	AF3
BLEU-1	0.277	0.251	0.368	**0.407**	0.640	−2.49	−2.55	−1.67	−1.40
BLEU-4	0.004	0.002	0.015	**0.030**	0.234	−1.27	−1.28	−1.18	−1.05
METEOR	0.124	0.102	0.151	**0.174**	0.319	−1.84	−2.05	−1.57	−1.32
CIDEr-D	0.048	0.064	0.164	**0.191**	0.906	−1.83	−1.75	−1.54	−1.41
SPICE	0.067	0.051	0.091	**0.117**	0.232	−1.47	−1.58	−1.19	−0.91
SPIDEr	0.057	0.058	0.127	**0.154**	0.569	−1.90	−1.84	−1.59	−1.41
FENSE	0.401	0.387	0.474	**0.510**	0.575	−1.10	−1.05	−0.73	−0.48
BERTScore-F1 (rescaled)	0.254	0.238	0.331	**0.352**	0.598	−2.40	−2.47	−1.80	−1.51
CLAP similarity	0.466	0.381	0.462	**0.509**	0.503	−0.34	−0.94	−0.42	+0.06†
MACE	0.573	0.489	0.575	**0.608**	0.635	−0.61	−1.18	−0.69	−0.34
Caption length (words)	21.0	16.4	17.0	19.0	11.3	+2.40	+1.30	+1.13	+1.16

Table 2. Review of 50 sampled captions per model, first against the references and then against the audio. Counts out of 50; last row: inaudible phrases among those no reference supports.

	GPT	Gemini	Qwen	AF3
Against the references				
Categories 1–3 (agree)	15	8	21	**26**
Category 4 (missing detail)	10	6	5	5
Category 5 (added detail)	10	12	9	9
Category 6 (weak or mismatch)	15	24	15	10
Against the audio				
Rated accurate	**47**	23	25	27
Minor errors	3	18	20	16
Major error	0	9	5	7
Category 6 and major error	0	6	1	1
Unsupported phrases	1/59	34/80	20/58	19/65

audio ratings, concordance ranges from 0.39 (rescaled BERTScore) to 0.58 (MACE, 95% CI [0.46, 0.69]; CLAP similarity 0.56 [0.45, 0.66]), and no metric differs significantly from chance after Holm correction. The intervals are wide and compatible with both chance and useful discrimination.

5. DISCUSSION

The results separate two properties that reference-based evaluation treats as one: agreement with the references and correctness about the audio. The text metrics agree with each other, with the reference-only review and with the model order of RQ1, but none of them showed evidence of following the audio review across models or within recordings. GPT-Audio-1.5 is the clearest case: it ranks third or fourth on every text metric, yet its captions were rated most accurate, because its deviations from the references are omissions and correct additional details rather than errors. The models' longer captions and the negative length–score correlations suggest that part of the gap to the human baseline reflects the short, generic style of Clotho's references rather than wrong content. We did not control for length, so this explanation remains untested.

CLAP similarity, which compares the caption with the audio, is the only metric on which a model is not significantly below the human baseline, and it comes closest to the audio-review order. For LALM captions we therefore recommend reporting audio-aware metrics alongside reference-based ones and checking a sample of captions against the audio before reading score differences as quality differences.

6. LIMITATIONS

The audio ratings are single-reviewer judgements on 50 recordings, without an independent second listener; the reviewer had read the references before listening, and the order of Audio Flamingo 3, Qwen2.5-Omni-7B and Gemini 3.8 Flash in the audio review is not stable (33% of resamples). The concordance analysis has little power (156 pairs from 42 recordings), so the absence of evidence that metrics track correctness is not evidence that they cannot. Each model was run with one prompt and one decoding setting, and the training data of the API models are undisclosed, so overlap with Clotho cannot be ruled out. Finally, the findings are limited to Clotho's everyday and environmental sounds. Future work should collect audio ratings from several independent listeners who have not seen the references, report inter-rater agreement, and use a larger sample to estimate concordance more precisely.

7. CONCLUSION

On Clotho v2.1, the open-weight LALMs agree most with the references (RQ1), but every model remains below a matched human baseline on every reference-based metric and writes captions 1.4–1.9 times longer than humans (RQ2). Most captions that disagree with the references had no major error against the audio, most details that no reference supports are audible, and the model ranked third or fourth by every text metric was rated most accurate (RQ3). Reference agreement alone is therefore insufficient evidence of acoustic correctness; claims about acoustic quality need evidence from the audio. Code and data: `https://github.com/anandaswarup/lalm-acoustic-caption-eval`.

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9. COMPLIANCE WITH ETHICAL STANDARDS

This study uses the publicly available Clotho dataset and captions generated by AI models. No human participants were recruited, and no ethical approval was required.

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